

THE AI TOOLBOX · CLASS 14

Experiment Design & Analysis

Turn a model score into evidence someone else can trust.

Merit Point Academy · 60 min · 8 quick checks · 2 runnable labs

RECAP · WHERE WE LEFT OFF

Three models are not yet an experiment

KNN

Votes with nearby examples; scaling changes who counts as nearby.

SVM

Protects a wide margin; C controls the trade.

Neural network

Learns layered features; seed, scaling and training choices matter.

Today's question: when their scores differ, how do you know the model caused the difference?

One claim, built in layers

1. **Question + baseline** — what must a model beat?
2. **Controlled comparison** — change one thing; freeze the rest
3. **Leakage + same split** — keep tomorrow out of training
4. **Cross-validation** — mean $\pm$ spread, not one lucky score
5. **Confusion matrix** — name both kinds of error
6. **Precision, recall, F1** — choose the metric before seeing results
7. **Threshold** — connect mistakes to real costs
8. **Two labs** — reproduce every displayed number

Eight quick checks are spaced through the hour. Pause, choose, then explain the answer before moving on.

WARM-UP · FAIR OR NOT?

Quick check · tap an answer

Teaching example: Model A scores 0.91 and Model B scores 0.86. A used one test split; B used a different split. What may you conclude?

A is better by 0.05

B is more stable

Nothing about the models yet — the test rows changed too

Average the two scores and pick A

DESIGN STEP 1 · WRITE THE CLAIM FIRST

A fair experiment begins as one sentence

On the same data, split, preprocessing and metric, does model B improve F1 over the majority-class baseline?

Independent variable

The one thing you change: model, feature set, or parameter.

Outcome

The metric chosen before results: F1 here.

Controls

Rows, folds, preprocessing, seed and evaluation code.

Every model must beat a baseline

A baseline is the simplest credible rule that answers the same question.

- Classification: always predict the majority class
- Regression: predict the training mean
- Time series: repeat the last observed value

Seeded synthetic teaching data

Majority baseline: accuracy **0.713**, recall **0.000**, F1 **0.000**.

It gets 107 / 150 rows right while finding **zero** positive cases.

High accuracy can be the lazy answer. The baseline tells you how much accuracy was available without learning anything useful.

Quick check · tap an answer

Teaching example: a dataset is 72% class 0. An always-class-0 baseline gets 72% accuracy and 0% recall for class 1. Your model gets 74% accuracy and 5% recall. Best interpretation?

The model is excellent because it beats baseline

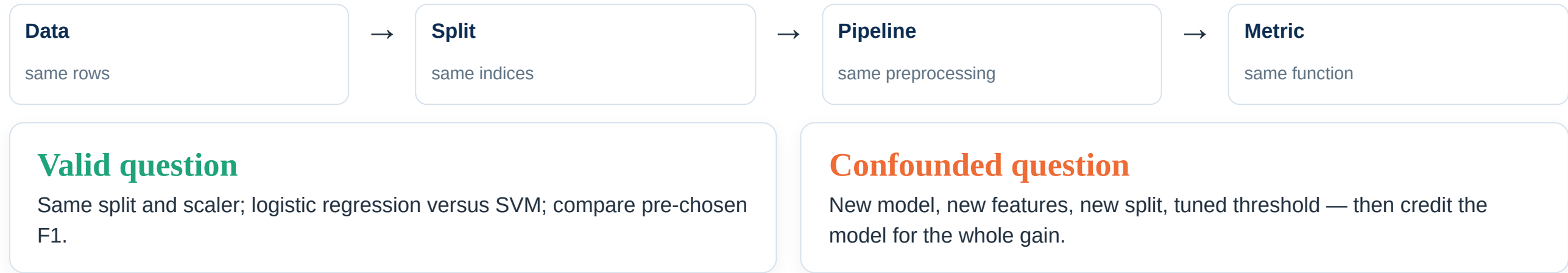
The model improves slightly, but the gain may be practically weak and needs uncertainty

Recall never matters if accuracy rises

Delete the baseline from the report

DESIGN STEP 3 · CONTROLLED COMPARISON

Change one thing; freeze the rest



The split is part of the experiment

The same logistic model, evaluated on 20 valid stratified splits, produced:

F1 min **0.563** · max **0.737**
mean **0.631** ± 0.050

Seeded synthetic teaching data · computed by Lab 1.

The fair pattern

1. Create one split or fold object
2. Save the indices
3. Pass the same indices to every contender
4. Never keep rerolling until one model wins

The split can move F1 by 0.174. That is larger than many claimed model improvements.

QUICK CHECK 3 · SAME SPLIT

Quick check · [tap an answer](#)

Teaching example: KNN uses random_state=7 and SVM uses random_state=42. SVM wins by 0.04 F1. What is the correct next move?

Publish SVM as the winner

Give KNN a larger k

Run both on the same saved split or folds

Subtract 0.04 from both scores

FAILURE MODE · TOMORROW SNEAKS INTO TRAINING

Leakage makes a model look clairvoyant

Target leakage

A feature is created from the answer or from information available only after the decision.

Train/test contamination

Scaling, imputation, feature selection or duplicate removal sees test rows.

Group/time leakage

The same person appears in both sets, or future events train a past prediction.

Synthetic demonstration: honest five-fold accuracy 0.810; add a near-copy of the label as a feature and it becomes 1.000. The perfect score is the alarm.

Preprocessing belongs inside the pipeline

Wrong

```
X_scaled = scaler.fit_transform(X) # all rows
cross_val_score(model, X_scaled, y)
```

The test fold helped choose its own means and standard deviations.

Right

```
pipe = make_pipeline(StandardScaler(), model)
cross_val_score(pipe, X, y, cv=folds)
```

Each fold learns preprocessing from its training rows only.

Split first. Fit transformations second. Apply them to held-out rows third.

QUICK CHECK 4 · LEAKAGE

Quick check · tap an answer

Before cross-validation, you fill every missing value using the median computed from the full dataset. What is wrong?

Nothing — medians cannot leak

The held-out fold influenced its own preprocessing

Cross-validation only works without missing values

Use the test-fold median instead

ANALYSIS STEP 1 · USE EVERY ROW HONESTLY

Cross-validation asks five honest versions

Fold 1

train on 4 · test on 1

→

Fold 2

train on 4 · test on 1

→

Fold 3

train on 4 · test on 1

→

Fold 4

train on 4 · test on 1

→

Fold 5

train on 4 · test on 1

StratifiedKFold

Preserves class proportions in each fold. Use for classification when rows are independent.

Group / time-aware folds

Keep each person in one fold; keep future rows after past rows. The split must match deployment.

Mean alone hides the unstable fold

Five F1 scores

0.727 · 0.610 · 0.710 · 0.596 · 0.448

$F1 = 0.618 \pm 0.112$

Sample standard deviation across fixed folds · synthetic teaching data.

- **Mean** describes typical performance
- **Standard deviation** shows sensitivity to which rows were held out
- **Fold scores** reveal whether one group is much harder

Do not write “the model scores 0.618” as if it were a law. Write the evaluation design and the spread beside it.

Quick check · tap an answer

Teaching example: Model A has $F1\ 0.72 \pm 0.14$; Model B has $F1\ 0.70 \pm 0.02$ on the same folds. Which is the strongest conclusion?

A is definitely better

B is definitely better

A has a slightly higher mean but much greater fold-to-fold variability

Standard deviation can be ignored when means differ

The confusion matrix names both mistakes

```
[[ TN 97, FP 10 ],  
 [ FN 16, TP 27 ]]
```

Seeded holdout from Lab 1:

- **16 false negatives** — positive cases missed
- **10 false positives** — false alarms
- Every other metric is a summary of these four boxes

Always state which class is “positive.” Swapping it changes precision and recall without changing accuracy.

Accuracy, precision, recall and F1 ask different questions

Accuracy 0.827 — how often was any label right?

Precision 0.730 — when it predicted positive, how often was it right?

Recall 0.628 — of all positives, how many did it find?

F1 0.675 — harmonic balance of precision and recall.

Choose before looking: false negatives costly → emphasize recall; false positives costly → emphasize precision; both matter → report both and perhaps F1.

QUICK CHECK 6 · METRIC CHOICE

Quick check · tap an answer

A screening-style task says missing a positive is far costlier than reviewing an extra false alarm. Which metric deserves priority?

Recall

Precision

Training accuracy

R-squared

ANALYSIS STEP 5 · THE MODEL GIVES A SCORE; YOU CHOOSE A RULE

The threshold moves the mistakes

threshold	precision	recall	F1	FP	FN
0.30	0.589	0.737	0.655	88	45
0.50	0.718	0.550	0.623	37	77
0.70	0.781	0.333	0.467	16	114

Out-of-fold probabilities · same five folds · synthetic teaching data.

Lower threshold: more positives called → recall rises, false alarms rise. Higher threshold: precision rises, misses rise. Retraining is not required.

DECISION RULE · WRITE COST BEFORE THRESHOLD

A threshold is a value judgment written as a number

Before seeing the table

Write: “A false negative costs five times a false positive.” Decide who owns that statement.

Then evaluate

For each threshold: $\text{cost} = 5 \times \text{FN} + 1 \times \text{FP}$. Show the full matrix so others can challenge the weights.

The 5:1 cost ratio is a teaching example, not a domain recommendation.

The model estimates probabilities. The project decides what to do with them. Do not hide the second decision inside “the AI chose.”

QUICK CHECK 7 · THRESHOLD

Quick check · tap an answer

**You lower a classifier's threshold from 0.70 to 0.30.
What usually happens?**

Precision and recall both always rise

Recall rises and false positives usually rise

The training data changes

The model's coefficients are retrained

Baseline vs. model — same split, same metric

Create one seeded synthetic dataset and one saved split. Compare a majority baseline with scaled logistic regression, then measure how much a rerolled split can move F1.

```
1 from sklearn.datasets import make_classification
2 from sklearn.dummy import DummyClassifier
3 from sklearn.linear_model import LogisticRegression
4 from sklearn.metrics import (accuracy_score, precision_score,
5                             recall_score, f1_score, confusion_matrix)
6 from sklearn.model_selection import train_test_split
7 from sklearn.pipeline import make_pipeline
8 from sklearn.preprocessing import StandardScaler
9
10 # Synthetic teaching data – not a business or clinical result.
11 X, y = make_classification(
12     n_samples=600, n_features=8, n_informative=5, n_redundant=1
```



Run



Explain



Debug



Improve



Mira

Python loads on first Run

► Run the code to see the output here.

Cross-validation mean $\pm$ spread, then move the threshold

Put scaling inside a Pipeline, collect five fold scores, create honest out-of-fold probabilities, and compare three thresholds without retraining.

```
1 import numpy as np
2 from sklearn.datasets import make_classification
3 from sklearn.linear_model import LogisticRegression
4 from sklearn.metrics import (accuracy_score, precision_score,
5                             recall_score, f1_score, confusion_matrix)
6 from sklearn.model_selection import (StratifiedKFold, cross_validate,
7                                     cross_val_predict)
8 from sklearn.pipeline import make_pipeline
9 from sklearn.preprocessing import StandardScaler
10
11 # The identical synthetic teaching dataset from Lab 1.
12 X, y = make_classification(
```



Run



Explain



Debug



Improve



Mira

Python loads on first Run

► Run the code to see the output here.

Quick check · tap an answer

Which results sentence is strongest?

Our AI achieved 81% and is production-ready

Logistic regression is best

On seeded synthetic teaching data, five-fold F1 was 0.618 ± 0.112 ; results may not generalize beyond this design

The model proved the hypothesis

REPORTING · MAKE THE CLAIM REPRODUCIBLE

The one-page experiment record

Design

Question · hypothesis · baseline · independent variable · controls · primary metric.

Protocol

Dataset version · exclusions · split/folds · seed · preprocessing pipeline · model settings.

Results

Fold scores · mean $\pm$ std · confusion matrix · threshold · limitations · failed runs retained.

Minimum honest sentence: “On [data], using [split/folds], model B changed [pre-chosen metric] from baseline [x] to [y], with [spread]; this does not establish [limitation].”

Submit the evidence, then present it yourself

What to submit

- Your slide deck
- A notebook that passes **Runtime** → **Run all**
- Your **one-page experiment record**

Talk: maximum **5:00**; aim for **4:30** · maximum **6 slides including title** · then **2:00 teacher questions**.

Sunday group

Due Sun · Aug 23, 2026 · 12:00 PM ET

Presentations begin 1:00 PM ET.

Tuesday group

Due Tue · Aug 25, 2026 · 7:00 PM ET

Presentations begin 8:00 PM ET.

Both deadlines are exactly one hour before the presentation class.

Aim for 4:30 — spend the minute on evidence

1 · 0:30

Question + metric

One answerable question; name the pre-chosen metric and why it fits.

2 · 0:40

Data

Source, version, n, target, balance or range, and one honest visual.

3 · 0:50

Experiment

Baseline, contender, one changed variable, controls, folds and Pipeline.

4 · 1:00

Results

Full table, uncertainty, and the error evidence behind the headline.

5 · 0:50

Interpretation + limits

Supported claim, surprise or failure, limitation, next test.

6 · 0:40

Reproducibility + AI

Notebook/data links, settings, and an AI-use verification table.

Hard limits: six slides includes the title · the talk stops at 5:00 · the following two minutes belong to teacher questions.

CLASS 15 · EVIDENCE CHECKLIST

If it is not visible, the audience cannot audit it

1. **Source, version and n**
2. Target + class balance or target range
3. Baseline + contender
4. One changed variable + **at least 3 controls**
5. Split/folds, seed and preprocessing Pipeline
6. Pre-chosen metric + why it matches the stakes

7. Full results table + uncertainty
8. For classification: confusion matrix; threshold + cost rule if moved
9. One surprise or failed result
10. One limitation + one next test
11. Notebook and data links
12. **AI table:** tool/model · used for · how I verified it

AI disclosure: write “No generative AI used,” or name every tool/model, what it helped produce, and the rerun, source check, or correction that made the work yours.

Be ready to defend the chain of custody

Teacher asks at least two

- What changed, and what stayed fixed?
- Why is this the right metric for the error costs?
- What is the largest threat to generalization?
- What evidence would make you reverse the conclusion?
- What AI suggestion did you reject, correct, or independently verify?

Peer feedback

Glow — strongest supported claim

Grow — one unclear evidence link

Next test — one experiment that could challenge the claim

Rubric · 100 points

Question + bounded claim	15
Data	10
Experiment fairness	20
Results + uncertainty	20
Interpretation + limits	15
Reproducibility + AI	10
Slides + timing	5
Q&A	5

WRAP-UP

A trustworthy score has a chain of custody

- **Baseline** — earn the complexity
- **Control** — same rows, pipeline, folds and metric
- **No leakage** — held-out means unseen
- **Uncertainty** — folds, mean and spread
- **Error costs** — matrix, metrics and threshold
- **Reporting** — say what the evidence cannot prove

Exit ticket: name the one variable your experiment changed — and three things it froze.

Next · Class 15 is your presentation class. You speak; the evidence answers back.